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EPIDEMIOLOGY

Superparamagnetic iron oxide (SPIO) for axillary mapping in patients with ductal carcinoma in situ undergoing mastectomy: single-institution experience

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Abstract

Purpose Unnecessary axillary surgery can potentially be avoided in patients with DCIS undergoing mastectomy. Current guidelines recommend upfront sentinel lymph node biopsy during the index operation due to the potential of upstaging to invasive cancer. This study reviews a single institution's experience with de-escalating axillary surgery using superparamagnetic iron oxide dye for axillary mapping in patients undergoing mastectomy for DCIS.

Methods This is a retrospective single-institution cross-sectional study. All medical records of patients who underwent mastectomy for a diagnosis of DCIS from August 2021 to January 2023 were reviewed and patients who had SPIO injected at the time of the index mastectomy were included in the study. Descriptive statistics of demographics, clinical information, pathology results, and interval sentinel lymph node biopsy were performed.

Results A total of 41 participants underwent 45 mastectomies for DCIS. The median age of the participants was 58 years (IQR = 17; range 25 to 76 years), and the majority of participants were female (97.8%). The most common indication for mastectomy was diffuse extent of disease (31.7%). On final pathology, 75.6% (34/45) of mastectomy specimens had DCIS without any type of invasion and 15.6% (7/45) had invasive cancer. Of the 7 cases with upgrade to invasive disease, 2 (28.6%) of them underwent interval sentinel lymph node biopsy. All sentinel lymph nodes biopsied were negative for cancer.

Conclusion The use of superparamagnetic iron oxide dye can prevent unnecessary axillary surgery in patients with DCIS undergoing mastectomy.

Keywords Superparamagnetic iron oxide · SPIO · Sentinel lymph node biopsy · Delayed sentinel lymph node biopsy · Ductal carcinoma insitu · DCIS

Introduction

The incidence of ductal carcinoma in situ (DCIS) is on the rise [1]. In the US alone, the incidence of DCIS rose from 5.8 per 100,000 in 1975 to 32.5 per 100,000 in 2004. Similar trends have been observed in Europe and other parts of the world where screening mammograms and digital mammography are commonplace [2, 3]. With this rise in diagnosis comes a concomitant increase in surgical intervention, as surgery is the current mainstay of treatment [4, 5].

Most patients with DCIS undergo breast conservation surgery (BCS) followed by adjuvant radiation [6]. In these cases, routine sentinel lymph node biopsy is not performed unless there is an upgrade to invasive breast cancer on final surgical pathology as preinvasive cells usually do not metastasize [2–6]. Mastectomy is another surgical option for patients who are not good candidates for BCS, such as those patients with diffuse multifocal disease, inability to attain a good cosmetic outcome with breast conservation therapy, inability to undergo adjuvant radiation, and in cases of pathogenic genetic mutation where patients may opt for mastectomy or due to patient's preference [4, 5, 7]. Current guidelines recommend axillary sentinel lymph node biopsy (SLNB) in the setting of mastectomy since a delayed SLNB

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after mastectomy is often challenging due to difficulty in axillary mapping [8].

The importance of the omission of unnecessary axillary surgery and its consequent associated co-morbidities in the setting of DCIS has been reinforced in several national guidelines [4, 8–10]. The National Comprehensive Cancer Network (NCCN) and the American Society of Clinical Oncology (ASCO) guidelines recommend that for patients undergoing mastectomy for DCIS, SLNB should be considered given the risk of upgrade to invasive breast cancer. They, however, do not recommend routine SLNB in patients with DCIS undergoing BCS [4, 8]. Upgrade rates from DCIS to invasive breast cancer (IBC) have been reported to be 2 to 49% in some studies [11–13]. These estimates suggest that 50% to over 90% of patients who undergo surgery for DCIS could be spared axillary surgery. Recent findings from the Sentinot trial demonstrated that the novel technique of delayed SLNB is feasible in patients with DCIS when superparamagnetic iron oxide (SPIO) is used during the index operation [14]. This offers the opportunity for patients with pure DCIS to omit SLNB from their management, thereby sparing them the complications of axillary surgery.

Since the publication of the Sentinot trial, there have been no data regarding the experience of SPIO use outside of a clinical trial setting in this patient population. There are also limited data on the use of SPIO in DCIS patients undergoing mastectomy.

In this study, we outline our institution's 18 month experience of using SPIO for axillary mapping in patients with DCIS undergoing mastectomy.

Methods

A retrospective single-institution cross-sectional study was performed. The medical records of all patients who underwent mastectomy for a diagnosis of DCIS from August 2021 to January 2023 were reviewed.

The study inclusion criteria specified all patients who underwent mastectomy for DCIS who also had SPIO injected at the time of their index mastectomy. In our study, all patients had 2 ml of SPIO injected in the subareolar space and massaged for 5 min after the induction of anesthesia and prior to administration of pectoral nerve block. In previous studies, SPIO signals have been demonstrated in the axilla up to 4 weeks following injection into the breast [14, 15]. Patients who were found to have incidental DCIS in the contralateral breast were also included. Patients with a preoperative diagnosis of IBC or microinvasive cancer were excluded from the study. Clinical information including age, sex, diagnostic radiological data, indication for mastectomy, type of mastectomy and reconstruction, genetic test, number of days to SLNB since index mastectomy, multidisciplinary tumor board decision, core needle biopsy pathology, and final surgical pathology (including tumor size, grade, and receptor status) and BMI were obtained from patient records. Descriptive statistics were performed using StataCorp. 2021. Stata Statistical Software: Release 17. College Station, TX: StataCorp LLC. Institutional Review Board (IRB) approval was obtained for this study.

Results

A total of 41 patients underwent 45 mastectomies for DCIS. The median age of the patients was 58 years (IQR = 17; range 25 to 76 years), and the majority were female (97.6%). The most common indication for mastectomy was diffuse extent of disease (31.7%) followed by patient's choice (26.8%) and patients considered high-risk per family history or genetic test (19.5%). The majority of patients underwent reconstruction following mastectomy (71.1%). A pathogenic gene mutation for breast cancer (BRCA2, ATM, and CHEK2) was found in 7 (24.1%) of the 29 patients who underwent genetic testing. All patients underwent a pectoral nerve block prior to the procedure (Table 1).

On final pathology, 75.6% (34/45) of mastectomy specimens had pure DCIS without any type of invasion, 15.6% (7/45) had invasive cancer, and 8.9% (4/35) had microinvasive disease. On immunohistochemistry, 88.9% (40/45) were estrogen receptor (ER) positive, 71.1 (32/45) were progesterone receptor (PR) positive, and 11.1% (5/45) were both ER and PR negative.

The majority, representing 57.8% (26/45), had intermediate-grade DCIS followed by high-grade DCIS (40%). The majority of patients upgraded to invasive disease had high-grade DCIS on their initial biopsy (Table 2).

Of the 7 cases with upgrade to invasive disease, 2 (28.6%) of them underwent interval sentinel lymph node biopsy. Each case was reviewed by the treatment team and a multidisciplinary tumor board to discuss the role of sentinel lymph node biopsy in the patient's treatment. For the two patients who underwent interval SLNB, SPIO signal was easily detected on the skin surface in one patient and in the axilla in both patients. For the remaining five patients who did not undergo interval SLNB, four of them met the criteria for the American Board of Internal Medicine (ABIM) choosing wisely recommendations. The last patient was reviewed at a multidisciplinary tumor board and was considered low risk based on age and tumor biology and a decision was made to omit sentinel node biopsy (Table 3).

Discussion

Breast cancer is the second leading cause of cancer-related mortality among women in the USA. In the year 2023 alone, approximately 43,170 women are predicted to die of breast

Table 1 Patient characteristics

Total cohort, $N=41$	
Age, median	58 (IQR = 17, range 25–76)
Sex (%)	
Female	97.6
Race (%)	
White	82.9
Black	12.2
Others	4.9
BMI	27.7 (IQR = 6.9)
Insurance (%)	
Medicaid	2.4
Medicare	22
Private	75.6
Indication for mastectomy (%)	
Diffuse disease	31.7
Patient choice	26.8
High risk	19.5
Multifocal disease	12.2
Close to nipple	4.9
Previous radiation	2.4
Unknown	2.4
Mastectomies, $N=45$	
Type of mastectomy (%)	
Skin sparing	48.9
Total	28.9
Nipple sparing	22.2
Reconstruction	
Yes	71.1
No	28.9

Table 2 Summary of pathological findings

DCIS with invasion or microinvasion (%), $N=11$		Pure DCIS (%) $N=34$	Total, $N=45$ (%)
Grade (%)			
Low	–	2.9	2.2
Intermediate	27.3	67.7	57.8
High grade	72.7	29.4	40
Size of DCIS, mm (median)	60[a]	21.5	28
ER positive (%)	90.9	88.2	88.9
PR positive	54.5	76.5	71.1

[a]Median size of invasive group was 60 mm and median for microinvasion group was 21 mm

cancer in the USA [1]. DCIS, which is the earliest stage of breast cancer, comprises a fifth of all diagnosed breast cancers [1]. The surgical management of DCIS is similar to that of invasive breast cancer with the exception of axillary management and recommended resection margins [4, 8].

The current recommendation for the management of the axilla in women with early-stage and clinically node-negative invasive breast cancer is to perform SLNB unless they meet the requirements for ABIM choosing wisely recommendations for SLNB in which case SLNB can be omitted [4, 9]. For patients with DCIS, SLNB is generally reserved for patients whose final pathology is upgraded to invasive breast cancer after a lumpectomy or for those who undergo an upfront mastectomy [4].

Sentinel lymph node biopsy poses a significant risk of complications, such as lymphedema, wound infection, seroma, and pain [16]. The risk of lymphedema has been reported to be 5% in some studies for patients who undergo SLNB [16]. This increased risk of morbidity associated with SLNB has led to efforts at de-escalation and omission in situations where surgical nodal staging will not affect adjuvant therapy recommendations [10].

This study sought to establish that with the use of SPIO, a significant proportion of patients with DCIS undergoing mastectomy could be spared SLNB and its associated morbidity.

Previous studies have reported upgrade rates of DCIS to invasive cancer to range from 2 to 49% [11–13]. Many of these studies included all patients who underwent either a mastectomy or lumpectomy. In our study, the upgrade rate of DCIS to invasive breast cancer was 15.6%. Our study population was limited to patients who underwent mastectomy for DCIS, the majority of whom (65.8%) had diffuse or multifocal disease, previous radiation, or were considered high risk based on their family history or genetic mutations (Table 1).

The use of SPIO has been shown to enable lymphatic mapping in patients who require a delayed SLNB. This novel technique enables the omission of upfront SLNB in DCIS patients undergoing mastectomy if SPIO is injected for axillary mapping during the index mastectomy operation. In the event of an upgrade to IBC on final surgical pathology, it allows for a return to the operating room for a delayed SLNB. In our study, the use of SPIO during the index mastectomy operation enabled the omission of SLNB in all patients with pure DCIS (75.6%). The Society of Surgical Oncology (SSO), in line with the ABIM choosing wisely campaign, released a set of guidelines in 2016 discouraging the routine use of SLNB in women over the age of 70 with early-stage, clinically node-negative, hormone receptor-positive, and HER2-negative breast cancers in an effort to reduce unnecessary procedures that have no long-term survival benefits in this patient population [8, 9]. Adoption of the SSO choosing wisely recommendations has been varied. For instance, in a study published in the Annals of Surgical Oncology in the year 2020, 8 fellowship-trained breast surgeons, 7 surgical oncologists, and 3 general surgeons were interviewed regarding choosing wisely practices. The majority of the surgeons were unaware of the choosing wisely

Table 3 Characteristics of patients with incidental invasive breast cancer in mastectomy specimen

	Sentinel lymph node biopsy	No sentinel lymph node biopsy
Number of cases	2	5
Age/age range	< 60	69–73
Hormone receptor status	ER −/PR −/HER2 − and ER+/PR+/HER2 −	ER+/HER2 −
Average tumor size, mm (range)	5 (3–7)	3.6 (2.5–4.5)
Average number of nodes (range)	3 (2–4)	na
Decision process	Multidisciplinary tumor board	Choosing wisely recommendation[aa]

na not applicable

[aa]One patient aged 69 years with ER+/PR+/HER2 − was discussed at multidisciplinary tumor board and considered low risk for lymph node metastasis

recommendations for SLNB or the evidence supporting this guideline. Additionally, nearly all of the surgeons based their decision to perform SLNB on the patient's functional status rather than age [17]. There are no clear guidelines regarding sentinel lymph node biopsy for patients with microinvasive disease; due to the low rate of lymph node metastasis in this patient population [18], our institution does not routinely perform SLNB in patients with DCIS with microinvasion.

With the employment of choosing wisely and multidisciplinary team decision-making in our institution, an additional 20% of patients (8.9% with microinvasive and 11.1% with invasive breast cancer) were spared SLNB and its consequent morbidity.

The combination of SPIO use, choosing wisely, and multidisciplinary team decision-making led to the omission of SLNB in over 95% of our patients who underwent mastectomy for DCIS. Although it is unknown whether axillary mapping is reliable in the postmastectomy setting, some studies have noted some success with the use of traditional blue dye or radioactive colloid [19, 20]. However, until consistency in interval axillary mapping can be demonstrated with the use of blue and radioactive dye in this setting, SPIO may offer the opportunity for delayed SLNB in this patient population.

In our study, one patient who chose to undergo contralateral prophylactic mastectomy was found to have incidental DCIS in the prophylactic breast specimen on final pathology. This raises the question as to whether axillary mapping using SPIO could be employed in patients with high-risk lesions or high-risk patients undergoing prophylactic mastectomy since this would allow for interval SLNB in the event of an upgrade to IBC.

Currently, an ongoing randomized control trial, Delayed Sentinel Lymph Node Biopsy in Ductal Cancer in Situ (SENTINOT_2) trial, aims to investigate the use of SPIO nanoparticles as a tracer for delayed sentinel lymph node dissection in patients where upfront axillary surgery (SLND) is deemed unnecessary. This includes patients with a diagnosis of DCIS on core needle biopsy, an unclear BIRADS 4–5 planned for excisional biopsy, and women undergoing risk-reducing mastectomy. This trial is still accruing and would be expected to complete in December 2027 [21].

In our study, patients with IBC found on final pathology had a larger median lesion size (60 mm vs 21 mm) and a higher incidence of high-grade disease (72.7% vs 29.4%) compared to those with pure DCIS.

Limitations of our study include small sample size precluding statistical analysis for significance and association, and the limitations inherent in a retrospective single-institution study. We feel however there is value in this study as a proof of concept for the ongoing investigation of SPIO to further refine axillary management for those patients' undergoing mastectomy with a diagnosis of DCIS on biopsy. Future studies could be performed with a larger sample size to look into clinicopathologic factors that are significantly associated with the risk of upgrade and successful axillary mapping with SPIO.

Conclusion

The use of superparamagnetic iron oxide dye can prevent unnecessary axillary surgery in patients with DCIS undergoing mastectomy. Most patients operated on for DCIS did not have invasive disease. In most of those who did have invasive or microinvasive disease, the addition of sentinel node biopsy was not a factor in adjuvant decision-making. The use of SPIO and delayed sentinel node biopsy affords the opportunity to avoid unnecessary axillary surgery and is a valuable step forward in axillary management for patients undergoing mastectomy for DCIS. In this study, unnecessary axillary surgery was avoided in 95% (39/41) of patients with the use of SPIO and choosing wisely recommendations. choosing wisely recommendation was an important determinant of axillary management in our patient population. More data will be beneficial to evaluate the adoption, use, and performance of this new technology outside of a clinical trial setting.

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Data availability The datasets generated during and/or analyzed during the current study are not publicly available because of institutional privacy regulations but are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors have no conflicts of interest to declare.

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